SC-PaCMAP: Data-Adaptive Spatial-Continuity Dimension Reduction for Spatial Data

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Abstract

High-dimensional geochemical and geometallurgical data commonly contain nonlinear multivariate relationships and spatial dependence. Linear dimensionality-reduction methods cannot adequately represent nonlinear structure, while many nonlinear approaches favour either local neighbourhood preservation or global arrangement and do not explicitly account for spatial relationships between regionalised observations. This study introduces Spatial-Continuity PaCMAP (SC-PaCMAP), a spatially aware extension of PaCMAP that retains its balance between local and global structure while incorporating spatial continuity supported by the data. SC-PaCMAP preserves the original feature relationships learned by PaCMAP while introducing a separate spatial-continuity representation. Importantly, the method automatically estimates the strength of spatial continuity present in the multivariate data and adapts the spatial contribution accordingly, avoiding the need for an additional user-defined spatial-versus-feature trade-off parameter. Synthetic and real geochemical case studies demonstrate that SC-PaCMAP improves spatial coherence while retaining competitive multivariate delineation and the local–global structure-preservation advantages of PaCMAP. The method provides a data-adaptive and interpretable framework for spatially informed visualisation, pattern discovery, sample assessment, and exploratory geochemical or geometallurgical domaining.

Keywords: dimensionality reduction; spatial autocorrelation; Geary's C; geochemistry; geometallurgy; PaCMAP; spatial clustering

1. Introduction

Mine-site interpretation combines geochemical and mineralogical information with expected processing behaviour (Grunsky et al., 2023; Navarra et al., 2017). These datasets often contain many related variables rather than a single grade measure (Sadeghi et al., 2024). Multivariate analysis helps identify geochemical anomalies and separate samples with different material patterns (Sadeghi et al., 2024; Grunsky et al., 2023). It also supports representative sample selection and the definition of orebody domains for spatial modelling (Siddiqui et al., 2024; Fouedjio et al., 2017). However, direct interpretation becomes difficult when the data are high-dimensional, nonlinear, or spatially variable (Min et al., 2025; Riquelme and Ortiz, 2024).

Dimensionality reduction (DR) reduces this burden by representing multivariate observations in fewer dimensions while retaining selected structure from the original data (Shlens, 2014; Wang et al., 2021). In exploration geochemistry, DR is used to handle high-dimensional data, identify mineralisation patterns, and support visualization or clustering (Sadeghi et al., 2024; Min et al., 2025). It is also useful for representative sample selection, where coverage of a large multidimensional dataset can be difficult to judge directly (Siddiqui et al., 2024). However, a two-dimensional embedding is not geological truth, because DR can distort neighbourhoods or display apparent clusters that are absent from the original space (Wang et al., 2021).

DR methods can be linear or nonlinear (Min et al., 2025). PCA is a widely used linear method that re-expresses data through linear combinations of the original variables (Shlens, 2014). However, geological multivariate data can contain nonlinear correlations, heteroscedastic behaviour, and spatial trends (Riquelme and Ortiz, 2024). In the Baiyin copper case study, manifold-learning methods outperformed conventional factor analysis in capturing complex nonlinear geochemical patterns (Min et al., 2025). In a separate hyperspectral remote-sensing benchmark, nonlinear dimensionality reduction via regression outperformed linear PCA across reconstruction, atmospheric-variable estimation, and land-cover classification tasks (Laparra et al., 2015). Together, these geochemical and remote-sensing benchmarks justify testing nonlinear DR against linear baselines in geoscience applications (Min et al., 2025; Laparra et al., 2015).

Standard nonlinear DR is not automatically spatially aware; t-SNE and UMAP do not jointly model feature and spatial information in their standard forms (Zhou et al., 2025). In their standard formulations, UMAP and PaCMAP build low-dimensional relationships from the supplied high-dimensional data, without a separate geological-coordinate or spatial-autocorrelation component (McInnes et al., 2018; Wang et al., 2021; Zhou et al., 2025). This matters because spatial correlation is an important property of geological data and is not automatically handled by standard machine-learning models (da Silva and Boisvert, 2020). Geoscience case studies further show that explicitly modelling spatial dependence can improve machine-learning prediction over comparable non-spatial methods (Fouedjio and Talebi, 2022; Talebi et al., 2022). GWPCA, SpatialPCA, and SpaSNE show that location can provide useful structure that ordinary DR does not directly use (Harris et al., 2011; Shang and Zhou, 2022; Zhou et al., 2025). However, GWPCA is a linear local PCA method, while SpatialPCA and SpaSNE were developed for spatially resolved omics and use formulations different from PaCMAP's pairwise local-global objective (Harris et al., 2011; Wang et al., 2021; Shang and Zhou, 2022; Zhou et al., 2025).

Feature similarity and spatial contiguity are separate objectives in spatial clustering (Chavent et al., 2018). Increasing the spatial weight can improve contiguity while reducing feature-space quality, so the useful solution is a trade-off rather than the most spatially smooth partition (Chavent et al., 2018). SpatialPCA reports a related distinction: methods that recovered spatial domains well did not necessarily recover feature-defined cell types well, and vice versa (Shang and Zhou, 2022). Together, these results support evaluating spatial continuity and multivariate delineation separately rather than collapsing them into one target (Chavent et al., 2018; Shang and Zhou, 2022).

This balance makes PaCMAP a suitable starting point (Wang et al., 2021). PaCMAP was designed to preserve both local and global feature-space structure, addressing a central trade-off in nonlinear visualization (Wang et al., 2021). Its nearest-neighbour pairs preserve local relationships, while mid-near and further-point pairs help organize broader structure and separation (Wang et al., 2021). This gives the embedding both fine-scale neighbourhood detail and a broader arrangement of groups (Wang et al., 2021). A later LocalMAP paper also describes standard PaCMAP as a graph in high-dimensional space built from nearest-neighbour, mid-near, and further-point pairs (Wang et al., 2025). SC-PaCMAP keeps this feature graph and adds spatial refinement separately.

This study introduces Spatial-Continuity PaCMAP (SC-PaCMAP) to address this gap. The method keeps PaCMAP's complete feature graph $\mathcal{G}_F$ and constructs a separate spatial-continuity graph $\mathcal{G}_S$ from geographically close candidates. PCA provides joint multivariate directions, and Geary's contiguity ratio is computed on each direction to estimate positive short-range spatial autocorrelation over the geographic-neighbour graph (Geary, 1954; Shlens, 2014). Geography proposes candidate pairs, while similarity in the spatially continuous PCA directions determines which candidates receive spatial support. The target spatial share is estimated from the Geary-derived spatial strength and the unresolved attraction remaining in the feature and spatial graphs, avoiding a fixed spatial weight. Spatial attraction is introduced only during PaCMAP's final optimization phase, is ramped in gradually, and replaces an equivalent share of nearest-neighbour attraction. Edges occurring in both graphs are downweighted to limit duplicate attraction.

Accordingly, this study targets controlled spatial continuity rather than maximum spatial smoothness. The central research question is whether a separate, data-adaptive spatial graph can improve spatial continuity while preserving the local and global multivariate structure established by PaCMAP. The evaluation therefore reports spatial continuity and feature-space delineation separately, while geological labels are held out and used only as supporting evidence. SC-PaCMAP is intended as an exploratory geospatial machine-learning method for identifying patterns that are both spatially coherent and multivariately meaningful.

2. Materials and methods

2.1 Problem definition and method overview

Let $\mathbf{X} \in \mathbb{R}^{N \times P}$ be a numerical feature matrix and $\mathbf{S} \in \mathbb{R}^{N \times r}$ the associated coordinates, with $r \in \{2, 3\}$. The objective is a two-dimensional embedding $\mathbf{Y} \in \mathbb{R}^{N \times 2}$ that preserves the nonlinear feature structure represented by PaCMAP while giving controlled preference to locally continuous multivariate variation.

SC-PaCMAP retains PaCMAP's complete feature-neighbour, mid-near, and further-pair graphs, denoted $\mathcal{G}_F$, $\mathcal{G}_M$, and $\mathcal{G}_P$. Independently, it constructs a raw-coordinate candidate graph $\mathcal{G}_G$, estimates PCA-Geary continuity, and selects an additional spatial-continuity graph $\mathcal{G}_S$. The original first two PaCMAP phases are unchanged. At the start of Phase 3, the method estimates a target spatial-attraction share, ramps that share over 25 iterations, and transfers the same share out of feature-neighbour attraction. All geological and synthetic labels are excluded from preprocessing, graph construction, embedding, and parameter estimation.

2.2 Datasets and preparation

Table 1. Datasets and neighbourhood settings used in the reported experiments.

Dataset	N	P	Coord.	Classes	k	Evaluation range and role
Repeated-facies synthetic	1,024	50	XY	4	8	$K = 2, \ldots, 10$; known facies repeated across disconnected patches
CMiO	1,551	35	XYZ	7	10	$K = 2, \ldots, 10$; held-out deposit groups
North Batten	3,222	46	XYZ	15	10	$K = 2, \ldots, 15$; held-out lithologies from 22 drillholes

2.2.1 Synthetic repeated-facies benchmark

The controlled population was generated once with `default_rng(42)`. A 32×32 grid with nominal 25 m spacing received independent coordinate jitter with standard deviation approximately 1 m. The grid was divided into a 4×4 set of 8×8 patches. For patch indices (t_x, t_y), the facies was

$$c = (t_x + t_y) \bmod 4, \tag{1}$$

so each of the four facies occurred in four disconnected locations. The 50-dimensional feature matrix contained ten answer-bearing variables built from a balanced three-value facies signature, 20 nonlinear non-spatial confounders, and 20 independent Gaussian distractors. A successful method therefore had to use local continuity selectively while joining disconnected occurrences of the same facies.

2.2.2 Critical Minerals in Ores

The Critical Minerals in Ores (CMiO) table was obtained from the Geoscience Australia geochemistry database (Geoscience Australia, 2021). The source CSV contained 29,087 rows and 341 columns. Samples in UTM zone 54S were retained so that easting and northing shared one projected coordinate system. The existing `DEPOSIT_GROUP` field supplied the evaluation label. Thirty-five chemical variables were converted to numeric form and complete rows were retained.

The preparation workflow treated negative assay values as left-censoring codes. A value $-v$ was replaced by $v/2$, using the encoded magnitude when a companion limit was unavailable in the retained row. This is an explicit analytical assumption, not zero replacement or general missing-value imputation. Deposit-group classes with fewer than 20 retained observations were removed. The final z coordinate was negative interval-midpoint depth,

$$z_i = -\frac{\mathrm{TOP_DEPTH}_i + \mathrm{BASE_DEPTH}_i}{2}, \tag{2}$$

not absolute elevation. The resulting table contained 1,551 samples, 35 variables, 38 deposits, and seven deposit-group labels.

2.2.3 North Batten

North Batten combined public Northern Territory reporting packages from 2015, 2018, and 2019 (O'Rourke and Pietrass-Wong, 2015; Mortimer et al., 2018; Wallace et al., 2019). The source releases provided assay, collar, geology, survey, and lithology-code tables. Northern Territory templates were parsed from their `H1000` data headers. Only direct `ME-MS61` measurements shared by all three campaigns and present in at least 80% of chemistry-bearing rows in every campaign were eligible. Reported $< v$ results were converted to $v/2$, $> v$ results to v, and complete rows were then retained; no mean, median, nearest-neighbour, or model-based assay imputation was used.

A lithology was assigned when the largest overlap with a logged geology interval covered at least 50% of the assay interval. Non-rock codes were removed and final classes required at least 20 intervals. Collar coordinates used GDA94/MGA Zone 53. Sample midpoints were desurveyed by minimum curvature. Later collar elevations were unavailable or zero placeholders, so z represents collar-relative vertical displacement rather than Australian Height Datum elevation. Intervals beyond the final measured survey station were excluded. The final table contained 3,222 intervals, 46 variables, 22 drillholes, and 15 lithology classes.

2.3 Feature preprocessing

Each retained feature was standardized once over the complete descriptive benchmark population,

$$x_{ip}^{(z)} = \frac{x_{ip} - \mu_p}{\sigma_p}. \tag{3}$$

Coordinate-appended baselines standardized coordinates separately before appending them as ordinary features. SC-PaCMAP and SpaSNE received raw coordinates through a separate spatial pathway.

For the reported datasets, $P \le 100$, so the copied PaCMAP 0.9.1 path applied its global shift and scale followed by column centring. Let

$$m = \min_{i,p} x_{ip}^{(z)}, \qquad M = \max_{i,p} \left(x_{ip}^{(z)} - m \right). \tag{4}$$

The matrix used for PaCMAP pair construction was

$$x_{ip}^{*} = \frac{x_{ip}^{(z)} - m}{M} - \frac{1}{N} \sum_{a=1}^{N} \frac{x_{ap}^{(z)} - m}{M}. \tag{5}$$

The initial embedding was 0.01 times the first two PCA scores of this matrix.

2.4 Original PaCMAP backbone

For each anchor i, PaCMAP retrieves $\min(k + 50, N - 1)$ non-self Euclidean candidates. Let $d_{i(h)}$ be the hth candidate distance after self-removal and

$$\sigma_i = \max\left(\frac{d_{i(4)} + d_{i(5)} + d_{i(6)}}{3}, 10^{-10} \right). \tag{6}$$

Candidate j is ranked by

$$\delta_{ij} = \frac{\left\|\mathbf{x}_i^* - \mathbf{x}_j^*\right\|_2^2}{\sigma_i \sigma_j}. \tag{7}$$

The first k candidates form the directed feature-neighbour graph $\mathcal{G}_F$. Per anchor, $k_M = \mathrm{round}(0.5k)$ mid-near targets are generated by sampling six non-self points and retaining the second-nearest, while $k_P = \mathrm{round}(2k)$ further targets are sampled from non-neighbours (Wang et al., 2021).

Let

$$\widetilde{d}_{ij} = 1 + \left\|\mathbf{y}_i - \mathbf{y}_j\right\|_2^2. \tag{8}$$

The original PaCMAP objective is

$$\begin{aligned} L_P = w_F(t) \sum_{(i,j)\in\mathcal{G}_F} \frac{\widetilde{d}_{ij}}{10 + \widetilde{d}_{ij}} + w_M(t) \sum_{(i,j)\in\mathcal{G}_M} \frac{\widetilde{d}_{ij}}{10000 + \widetilde{d}_{ij}} \\ + w_P(t) \sum_{(i,j)\in\mathcal{G}_P} \frac{1}{1 + \widetilde{d}_{ij}}. \end{aligned} \tag{9}$$

The 450 optimization iterations use the schedule in table 2. Adam uses learning rate 1.0, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-7}$.

Table 2. PaCMAP optimization schedule retained by SC-PaCMAP.

Iterations	w_F	w_M	w_P
0–99	2	linearly 1000 → 3	1
100–199	3	3	1
200–449	1	0	1

2.5 Spatial-continuity graph

2.5.1 Geographic candidates and PCA-Geary continuity

Each observation receives its k nearest non-self candidates in raw coordinate space, forming the directed graph $\mathcal{G}_G$. Raw coordinates determine eligibility only; coordinate distance is not added to the feature distance.

A second PCA is fitted to the PaCMAP-preprocessed feature matrix. Let $Q = \min(100, P^*, N-1)$, r_{iq} be the score of sample i on component q, ν_q its explained variance, and π_q its explained-variance ratio. Scores are standardized to u_{iq} with population standard deviation. Component-wise Geary continuity on the directed geographic graph is (Geary, 1954)

$$C_q = \frac{(N-1)\sum_{i=1}^{N}\sum_{j\in\mathcal{G}_G(i)}(u_{iq} - u_{jq})^2}{2Nk\sum_{i=1}^{N}(u_{iq} - \bar{u}_q)^2}. \tag{10}$$

Non-finite values are set to one. The implementation uses

$$b_q = \mathrm{clip}(1 - C_q, -1, 1), \qquad a_q = \max(0, b_q). \tag{11}$$

The signed value b_q contributes to the global strength estimate, while only positive continuity a_q supports SC edges.

2.5.2 Adaptive edge supply and candidate ranking

The global spatial-strength estimate and number of additional SC edges per anchor are

$$\lambda = \mathrm{clip}\left(\sum_{q=1}^{Q}\pi_q b_q, 0, 1\right), \qquad k_S = \mathrm{int}(\mathrm{round}(k\lambda)). \tag{12}$$

The k original feature neighbours are retained in full. For geographic candidate $j \in \mathcal{G}_G(i)$, the spatial-continuity distance is

$$D_{ij}^{\mathrm{SC}} = \sum_{q=1}^{Q}\nu_q a_q (u_{iq} - u_{jq})^2. \tag{13}$$

The k_S smallest values per anchor are selected by stable sort to form $\mathcal{G}_S$. A selected edge is therefore geographically local and similar in multivariate directions that exhibit positive short-range continuity.

Let m_D be the median strictly positive selected distance, with fallback $m_D = 1$ and lower bound 10^{-12}. Raw edge confidence is

$$\widetilde{\omega}_{ij} = \frac{1}{1 + D_{ij}^{\mathrm{SC}}/m_D}, \qquad \omega_{ij}^{(0)} = \frac{\widetilde{\omega}_{ij}}{\mathrm{mean}_{(a,b)\in\mathcal{G}_S}\,\widetilde{\omega}_{ab}}. \tag{14}$$

If the exact directed pair (i, j) also belongs to $\mathcal{G}_F$, its SC weight is multiplied by 0.5 to limit duplicate attraction. Weights are not renormalized after this overlap correction.

2.6 Phase-3 spatial loss and attraction-budget transfer

SC attraction is absent in Phases 1 and 2. In Phase 3 it uses

$$L_{\mathrm{SC}}(t) = w_S(t) \sum_{(i,j)\in\mathcal{G}_S} \omega_{ij} \frac{\widetilde{d}_{ij}}{C_{\mathrm{SC}} + \widetilde{d}_{ij}}, \qquad C_{\mathrm{SC}} = 10. \tag{15}$$

At iteration $t_0 = 200$, the implementation measures unresolved attractive-force scales in the feature and SC graphs. Writing $r_{ij} = \left\|\mathbf{y}_i - \mathbf{y}_j\right\|_2$,

$$A_F(\mathbf{Y}) = \sum_{(i,j)\in\mathcal{G}_F} \frac{20 r_{ij}}{(10 + \widetilde{d}_{ij})^2}, \tag{16}$$

$$A_S(\mathbf{Y}) = \sum_{(i,j)\in\mathcal{G}_S} \omega_{ij} \frac{2 C_{\mathrm{SC}} r_{ij}}{(C_{\mathrm{SC}} + \widetilde{d}_{ij})^2}. \tag{17}$$

Their effective per-edge magnitudes are

$$\bar{A}_F = \frac{A_F}{|\mathcal{G}_F|}, \qquad \bar{A}_S = \frac{A_S}{\sum_{(i,j)\in\mathcal{G}_S} \omega_{ij}}. \tag{18}$$

With $\lambda_c = \mathrm{clip}(\lambda, 0, 0.95)$, the target spatial share is computed once as

$$\rho^* = \mathrm{clip}\left(\frac{\lambda_c \bar{A}_S}{\lambda_c \bar{A}_S + (1-\lambda_c)\bar{A}_F}, 0, 0.95\right). \tag{19}$$

An empty or degenerate SC graph gives $\rho^* = 0$. The target is introduced over the first 25 Phase-3 iterations,

$$\rho_t = \rho^* \,\mathrm{clip}\left(\frac{t - 200}{25}, 0, 1\right). \tag{20}$$

At every Phase-3 iteration, A_F and A_S are recomputed at the current embedding, while ρ^* remains fixed. The original Phase-3 feature-neighbour weight $w_F^{(0)} = 1$ is redistributed as

$$w_F(t) = w_F^{(0)}(1 - \rho_t), \qquad w_S(t) = w_F^{(0)} \rho_t \frac{A_F}{A_S}. \tag{21}$$

Thus $w_F A_F + w_S A_S = w_F^{(0)} A_F$ under the same scalar force proxy. Spatial attraction is transferred from the existing local-attraction budget rather than added as uncontrolled force. The mid-near and further-pair terms and the Adam update remain unchanged.

2.7 Benchmark methods and evaluation

The proposed method was compared with feature-only PaCMAP; PaCMAP, UMAP, and LocalMAP after separately standardized coordinates were appended to the features; SpaSNE with raw coordinates; and the earlier neighbour-replacement ablation, denoted SC-PaCMAP-NR. The ablation uses the same PCA-Geary candidate logic but replaces feature neighbours instead of retaining the full feature graph. It was included for the two real datasets and omitted from the synthetic six-method comparison.

PaCMAP-family methods used MN ratio = 0.5, FP ratio = 2.0, learning rate 1.0, seed 42, and (100, 100, 250) iterations. UMAP used the matched neighbour count, Euclidean distance, learning rate 1.0, and seed 42. LocalMAP used the same PaCMAP-family schedule and `low_dist_thres=10`. SpaSNE used $\alpha = 10$, $\beta = 5$, exaggeration 4, perplexity 50, $\theta = 0.5$, 1,000 iterations, PCA preprocessing, and seed 42.

Each raw two-dimensional embedding was clustered by K-means++ with 20 initializations and random state $1000 + K$.

Algorithm 1 SC-PaCMAP

Require: standardized features $\mathbf{X}$, raw coordinates $\mathbf{S}$, PaCMAP neighbour count k
Run original PaCMAP preprocessing and construct $\mathcal{G}_F, \mathcal{G}_M, \mathcal{G}_P$.
Construct raw-coordinate directed kNN candidates $\mathcal{G}_G$.
Fit PCA to preprocessed features; standardize scores; compute C_q, b_q, and a_q.
Compute λ and k_S; select the k_S smallest D^{SC}_{ij} within each $\mathcal{G}_G(i)$ to form $\mathcal{G}_S$.
Convert D^{SC} to mean-one confidence weights and halve directed overlaps with $\mathcal{G}_F$.
Initialize $\mathbf{Y}$ using original PaCMAP PCA initialization.
for $t = 0, \ldots, 449$ **do**
    Obtain original PaCMAP weights for iteration t.
    if $t < 200$ or $\mathcal{G}_S = \varnothing$ **then**
        Set $w_S = 0$ and use the original PaCMAP gradient.
    else
        Recompute A_F, A_S; at $t = 200$, compute and store ρ^*.
        Compute $\rho_t, w_F(t), w_S(t)$ using equations (20) and (21).
        Evaluate the original PaCMAP gradient with adjusted $w_F(t)$ and append the SC gradient.
    end if
    Apply the original Adam update.
end for
return Y

Labels were used only to define a descriptive reference K and to calculate adjusted Rand index (ARI) (Hubert and Arabie, 1985). The principal evaluation was deliberately two-objective.

First, spatial continuity was measured using the same raw-coordinate neighbour count as the embedding. If p_{ic} is the local proportion of cluster c among sample i's geographic neighbours, mean windowed entropy is

$$\frac{H}{N} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{K} p_{ic}\log p_{ic}. \tag{22}$$

Lower values indicate less local cluster mixing. Second, multivariate delineation was measured in the standardized original feature space,

$$\frac{\mathrm{WCSS}}{\mathrm{TSS}} = \frac{\sum_{c}\sum_{i:z_i=c}\|\mathbf{x}_i - \bar{\mathbf{x}}_c\|_2^2}{\sum_{i}\|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2}, \tag{23}$$

where lower values indicate tighter feature-space groups. Movement toward the lower-left of an entropy–WCSS plot improves both objectives. ARI was retained as evaluation-only supporting evidence.

3. Results

The final separate-graph implementation is denoted SC-PaCMAP; the earlier neighbour-replacement implementation is denoted SC-PaCMAP-NR. All reported embeddings use deterministic seed 42. Disabling the SC component reproduced the copied PaCMAP pair arrays and embedding exactly, with maximum absolute embedding difference equal to zero. The copied path also reproduced the installed PaCMAP 0.9.1 implementation exactly. Differences below can therefore be attributed to the added spatial graph and its Phase-3 attraction.

3.1 Data-adaptive spatial allocation

Table 3. Data-adaptive graph and attraction diagnostics for the final SC-PaCMAP implementation.

Dataset	λ	Median C	Feature edges	SC edges	Target share	Realized share	Overlap
Synthetic	0.1252	0.9994	8	1	0.1752	0.1667	0.1270
CMiO	0.4572	0.7350	10	5	0.5618	0.5348	0.3700
North Batten	0.5106	0.6627	10	5	0.5456	0.5193	0.5575

Feature edges are retained in full; SC edges are additional. Overlap is the directed fraction of SC edges already present in the feature graph. Target and realized shares refer to active Phase-3 nearest-neighbour versus SC attraction.

The estimated spatial contribution differed substantially among datasets (table 3). The synthetic benchmark produced $\lambda = 0.1252$ and median Geary $C = 0.9994$, indicating that most of its 50 variables were effectively non-spatial at the

evaluated scale. The method consequently retained all eight feature neighbours and added only one SC edge per sample. CMiO and North Batten produced stronger continuity estimates, $\lambda = 0.4572$ and 0.5106, and each added five SC edges while retaining all ten feature neighbours. The realized active attraction shares remained close to their automatic targets.

Table 4. Reference-partition benchmark results. Lower H/N and WCSS/TSS are better; higher ARI is better.

Dataset (K)	Method	H/N	WCSS/TSS	ARI
Synthetic (4)	UMAP + XY	0.9443	0.9172	0.1581
	PaCMAP + XY	0.8044	0.9112	0.3454
	LocalMAP + XY	0.7439	0.9070	0.4044
	PaCMAP feature-only	0.7803	0.9075	0.2895
	SC-PaCMAP	0.3600	**0.8580**	**0.8728**
	SpaSNE	**0.1253**	0.9751	0.1407
CMiO (7)	UMAP + XYZ	0.4873	0.5932	0.3608
	PaCMAP + XYZ	0.5375	0.5891	0.3343
	LocalMAP + XYZ	0.3371	0.5652	0.3108
	PaCMAP feature-only	0.6709	**0.5595**	0.2641
	SC-PaCMAP-NR	0.2574	0.6379	0.4926
	SC-PaCMAP	**0.1766**	0.6045	**0.5242**
	SpaSNE	0.3831	0.6021	0.4213
North Batten (15)	UMAP + XYZ	0.3101	0.7298	0.1121
	PaCMAP + XYZ	0.3635	0.6443	0.1054
	LocalMAP + XYZ	0.3367	0.6891	0.1036
	PaCMAP feature-only	0.5690	**0.5854**	**0.1246**
	SC-PaCMAP-NR	0.2558	0.6444	0.1181
	SC-PaCMAP	**0.2306**	0.6357	0.1230
	SpaSNE	0.4015	0.6578	0.1082

Bold values are best within each dataset and metric. The synthetic row has six methods because the neighbour-replacement ablation was evaluated only on the two real datasets. Labels were not used in any embedding or parameter allocation.

3.2 Synthetic repeated-facies benchmark

The synthetic population contained 1,024 observations in 16 physical patches but only four facies, each repeated at disconnected locations (figure 1). The ten answer-bearing variables formed a minority of the 50-dimensional input, while the remaining variables created nonlinear and independent nuisance structure. Geographic smoothness alone could therefore preserve patches without recovering the repeated facies.

The embeddings show the distinction directly (figure 2). SC-PaCMAP clearly separated the four facies. UMAP + XY and the ordinary PaCMAP variants spread the same facies across overlapping manifolds; LocalMAP + XY separated local pieces but fragmented each facies into several islands; SpaSNE compressed the facies into one mixed cloud.

At $K = 4$, SC-PaCMAP achieved $H/N = 0.3600$, WCSS/TSS = 0.8580, and ARI = 0.8728, giving the best feature compactness and facies recovery (table 4). The three-method ablation explains the mechanism: feature-only PaCMAP mixed clusters within the true patches, appending XY changed the mixing but still missed the repeated pattern, whereas SC-PaCMAP recovered the four facies across all 16 patches (figure 3).

SpaSNE attained lower entropy (0.1253) but poor feature compactness (0.9751) and ARI (0.1407), illustrating that spatial smoothness alone is insufficient. Across $K = 2, \ldots, 10$, SC-PaCMAP retained the highest ARI and remained below feature-only PaCMAP in both trade-off dimensions, forming the dominant lower-left trajectory (figure 4).

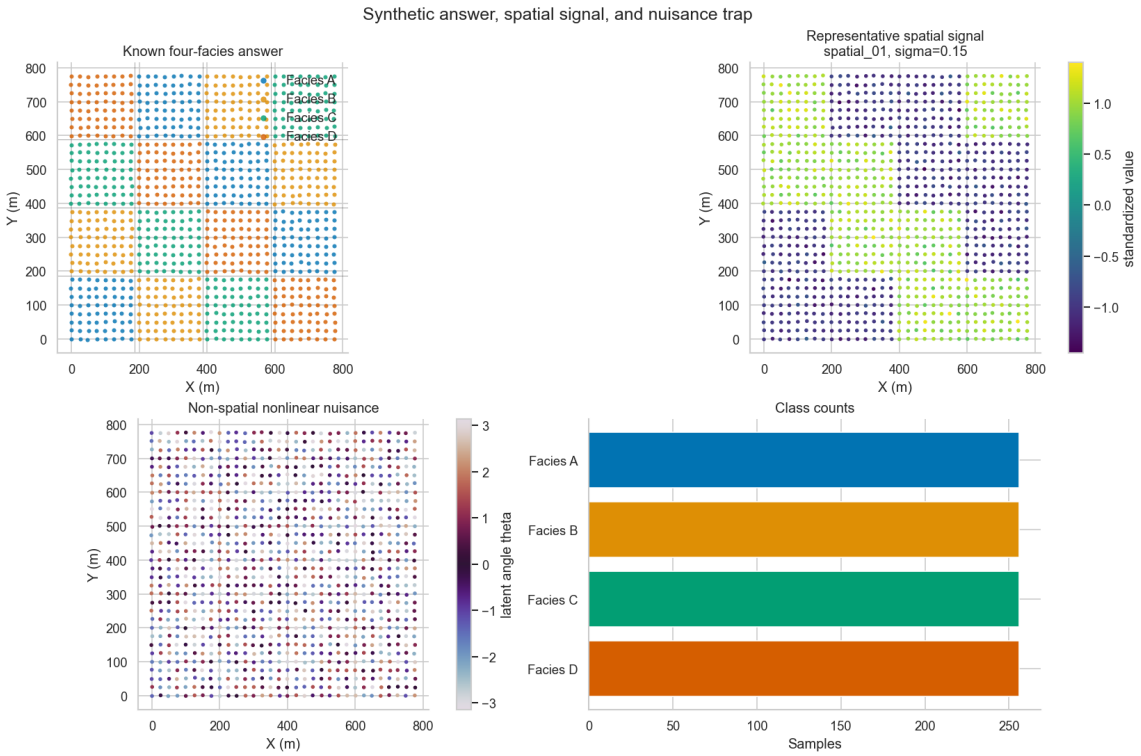


Figure 1. Controlled repeated-facies benchmark. Four facies repeat across 16 disconnected physical patches. A representative answer-bearing variable is locally continuous, whereas the nonlinear nuisance is deliberately non-spatial. All facies contain 256 observations.

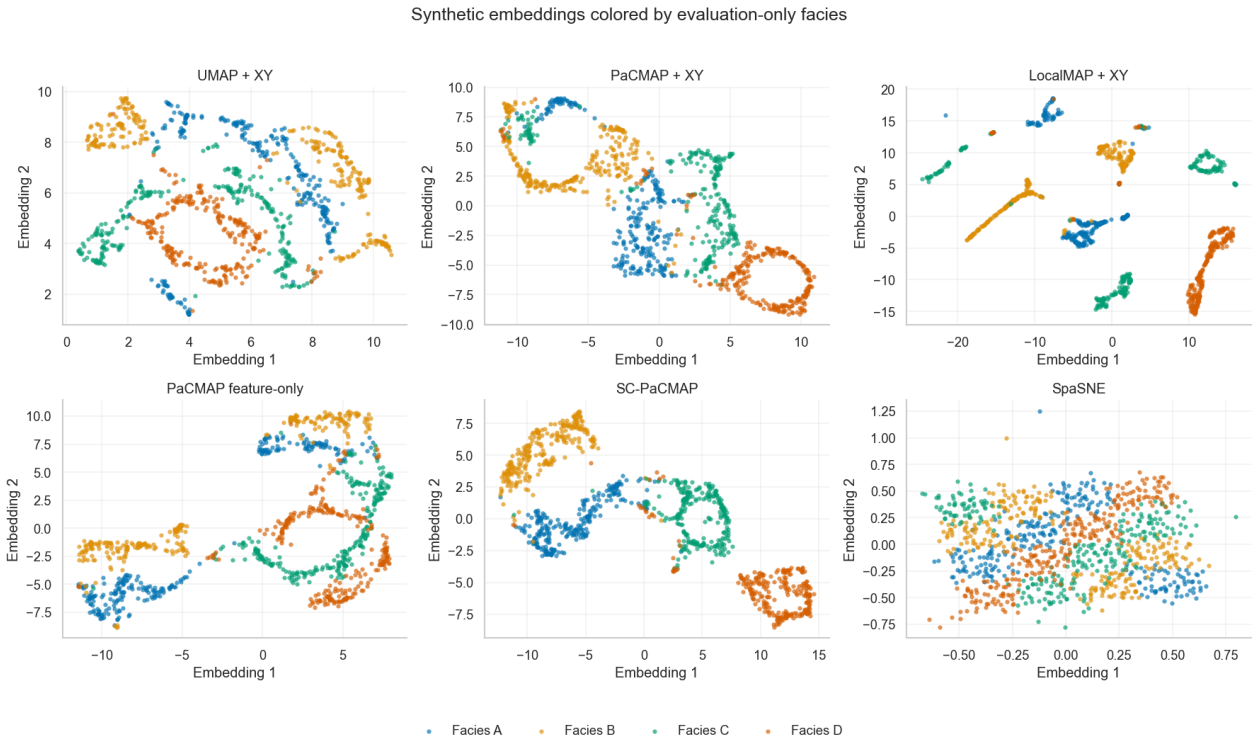


Figure 2. Six unsupervised embeddings coloured after fitting by held-out facies. SC-PaCMAP gives the clearest four-facies separation; LocalMAP fragments the facies, ordinary embeddings overlap them, and SpaSNE mixes them in a common cloud.

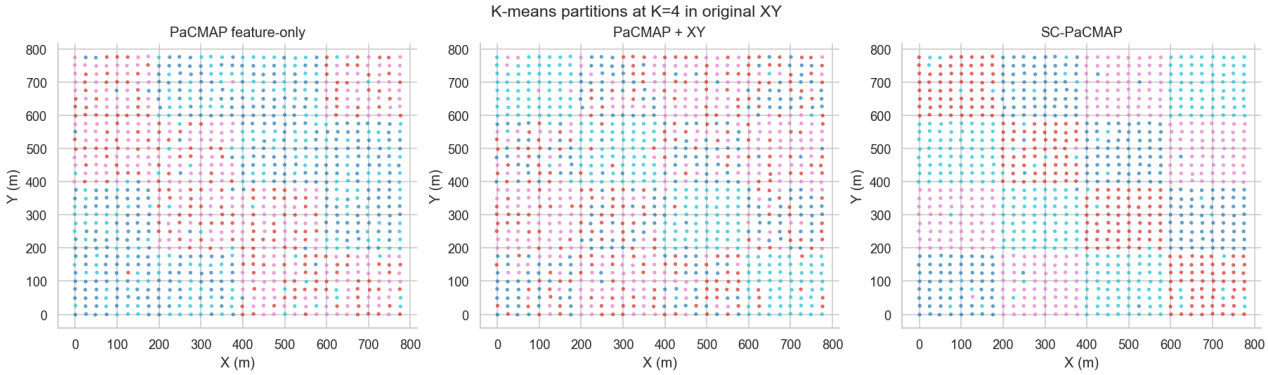


Figure 3. Ablation at $K = 4$ in original XY space. Feature-only PaCMAP and PaCMAP + XY mix the known patches; SC-PaCMAP recovers the repeated four-facies pattern.

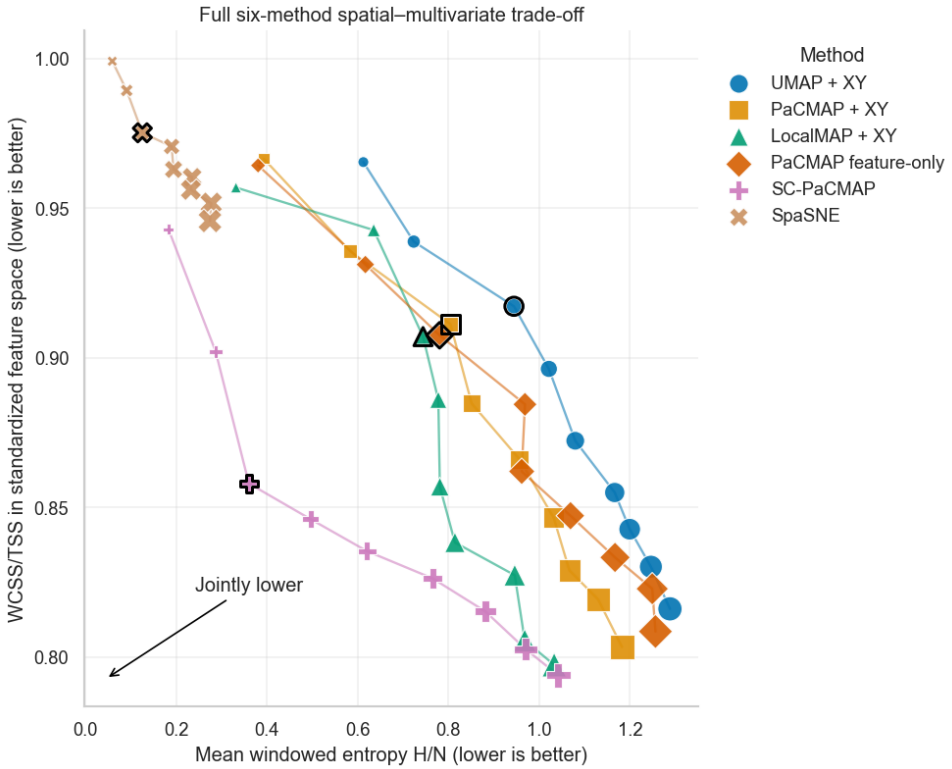


Figure 4. Synthetic spatial–multivariate trade-off across $K = 2, \ldots, 10$. Movement toward the lower left jointly improves spatial continuity and feature-space delineation; black outlines identify $K = 4$.

3.3 CMiO regional geochemical benchmark

CMiO contained 1,551 samples, 35 geochemical variables, XYZ coordinates, and seven held-out deposit-group labels. At $K = 7$, no method minimized every criterion. Feature-only PaCMAP gave the lowest WCSS/TSS across the compared methods (0.5595) but also the highest entropy (0.6709) and ARI 0.2641. SC-PaCMAP gave the lowest entropy (0.1766) and highest ARI (0.5242), with WCSS/TSS = 0.6045.

Relative to feature-only PaCMAP, SC-PaCMAP reduced entropy by 73.7% and increased ARI by 0.2601, with an 8.0% increase in WCSS/TSS. Against PaCMAP + XYZ, it reduced entropy by 67.1% and increased ARI by 0.1899 while increasing WCSS/TSS by only 2.6%. The result therefore cannot be reproduced by simply appending coordinates to the feature matrix.

The separate-graph design also improved on SC-PaCMAP-NR. At $K = 7$, SC-PaCMAP-NR obtained $H/N = 0.2574$, WCSS/TSS = 0.6379, and ARI 0.4926. Retaining the complete feature graph and adding a late spatial attraction reduced entropy by a further 31.4%, reduced WCSS/TSS by 5.2%, and increased ARI by 0.0316.

Across the full $K = 2, \ldots, 10$ sweep, SC-PaCMAP traced the low-entropy side of the candidate set and remained top or near-top in ARI (figure 5). Feature-only PaCMAP remained the feature-compact endpoint. In the joint plot, SC-PaCMAP formed the lower-left frontier from $K = 5$ onward while avoiding the weaker feature delineation of SC-PaCMAP-NR (figure 6). The largest ARI in the complete sweep was 0.6085 from SC-PaCMAP-NR at $K = 5$, so $K = 7$ should be interpreted as evaluation at the supplied deposit-group resolution, not an unsupervised optimum.

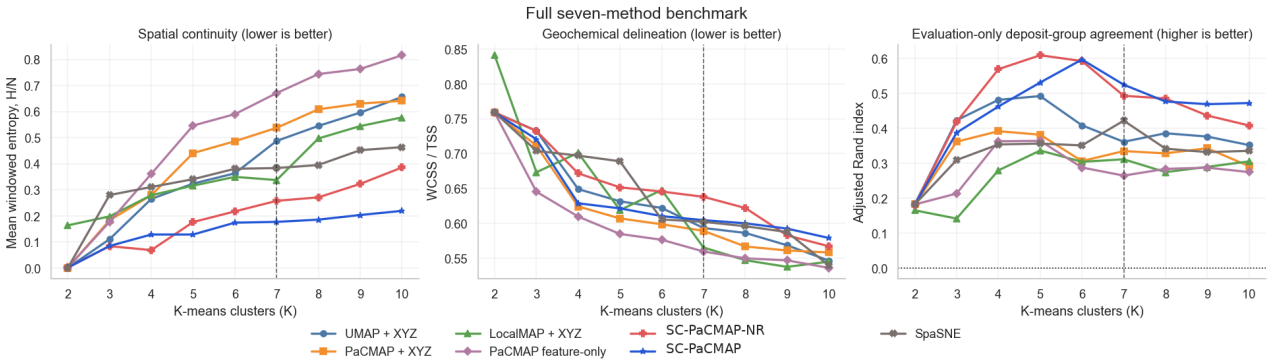


Figure 5. CMiO benchmark across $K = 2, \ldots, 10$: spatial continuity, geochemical delineation, and held-out deposit-group agreement. The dashed line marks $K = 7$. The final SC-PaCMAP method retains the complete feature graph; SC-PaCMAP-NR is the neighbour-replacement ablation.

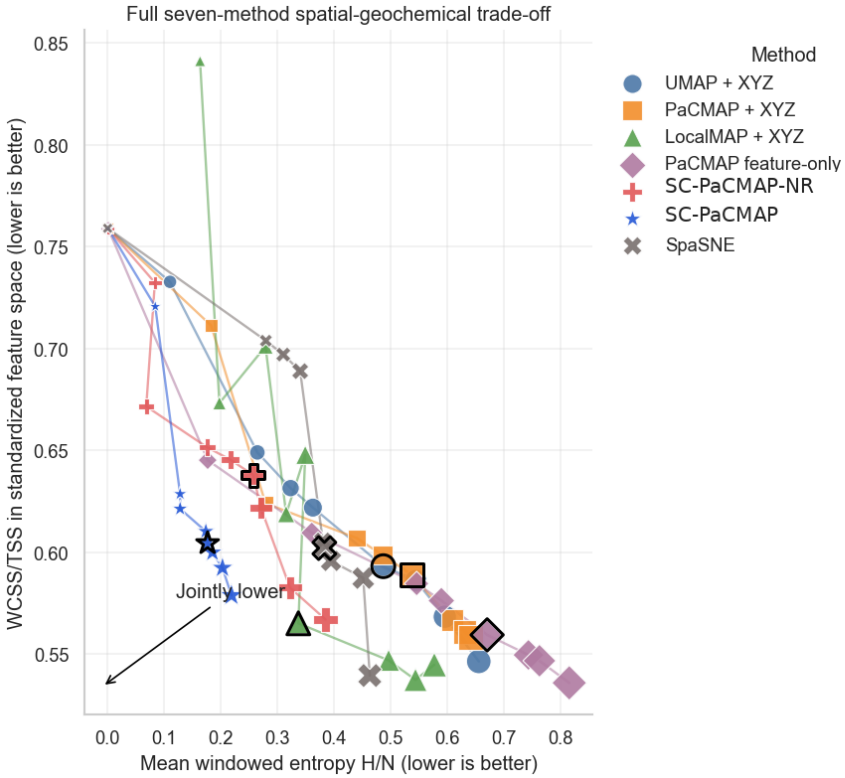


Figure 6. CMiO spatial–geochemical trade-off. Lower-left is preferred. SC-PaCMAP forms the joint frontier while feature-only PaCMAP remains feature-compact but spatially mixed. Black outlines mark the seven-group reference.

3.4 North Batten drill-core benchmark

North Batten contained 3,222 intervals, 46 geochemical variables, XYZ coordinates, 22 drillholes, and 15 held-out lithology classes. At $K = 15$, feature-only PaCMAP again provided the best WCSS/TSS (0.5854) but the worst entropy (0.5690). SC-PaCMAP reduced entropy to 0.2306 while retaining nearly identical class agreement (ARI 0.1230 versus 0.1246), with WCSS/TSS = 0.6357. It also improved all three reference metrics relative to SC-PaCMAP-NR and the coordinate-appended baselines except for the intentionally feature-focused PaCMAP endpoint.

At the reference partition, the trade-off plot places the final $K = 15$ solution close to feature-only PaCMAP in ARI while substantially reducing entropy (figure 7). Across the full sweep, feature-only PaCMAP stayed low in WCSS/TSS but far toward the spatially mixed side, whereas the two SC variants formed the low-entropy envelope (figure 8). The final method had lower WCSS/TSS than SC-PaCMAP-NR for $K = 6, \ldots, 15$. ARI was low for every method, so this case supports improved spatial–multivariate delineation rather than reconstruction of the logged lithology taxonomy.

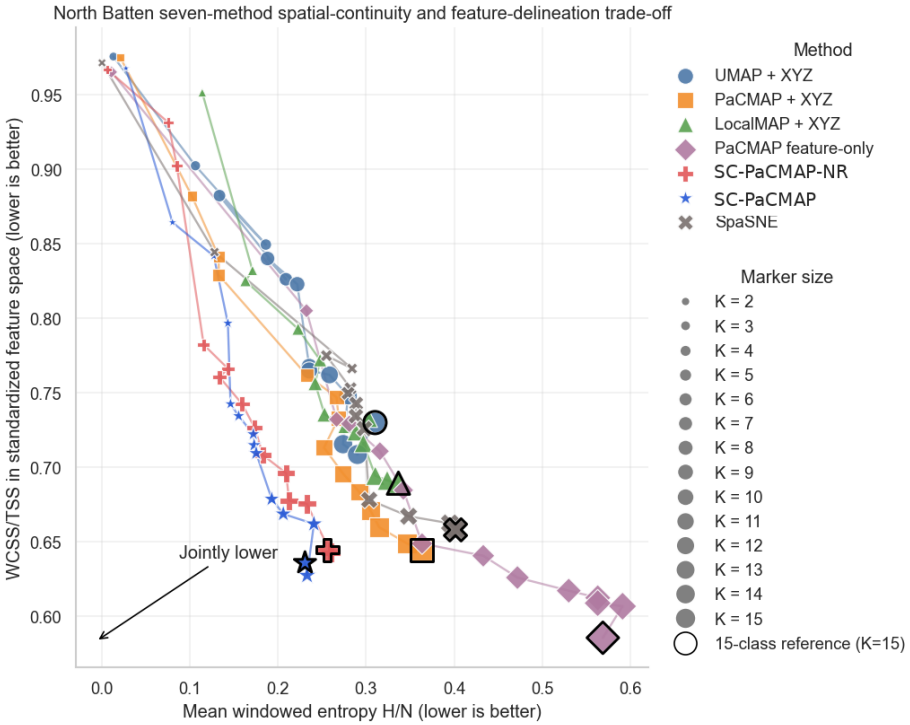
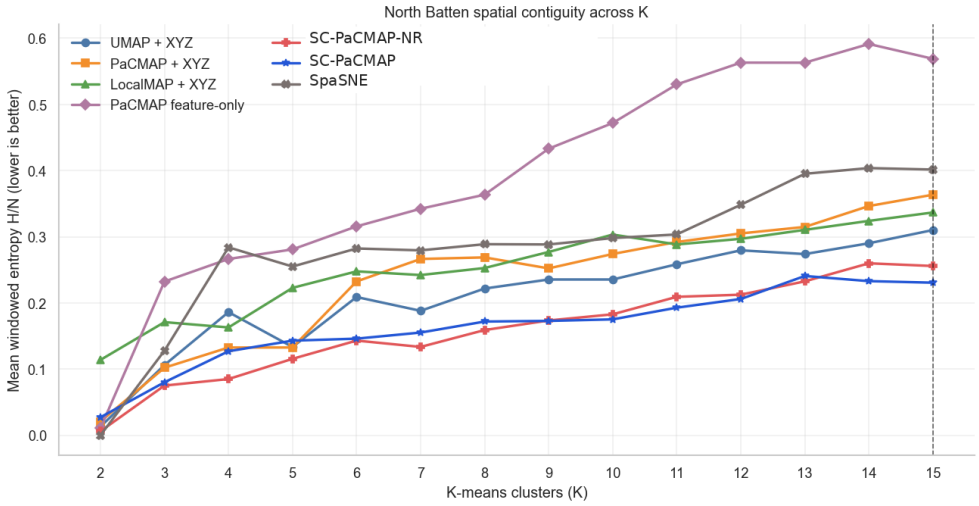
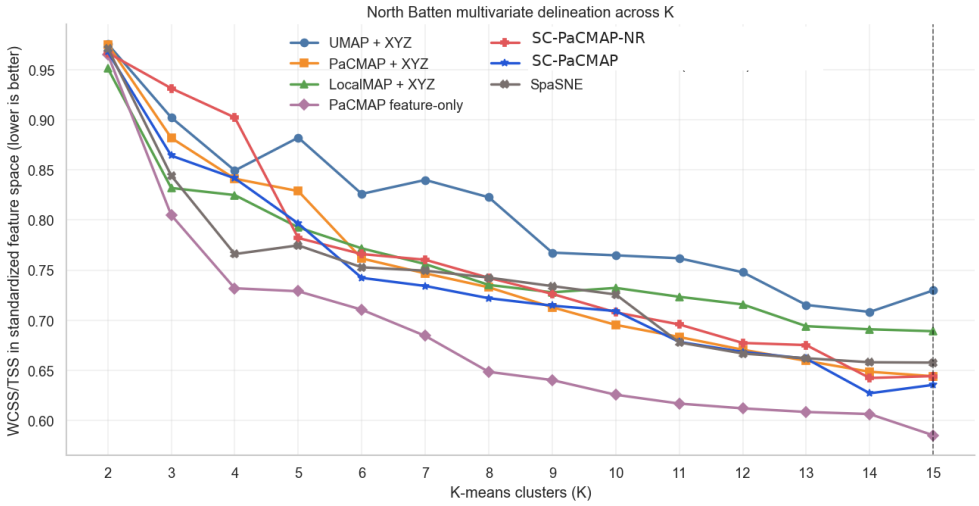


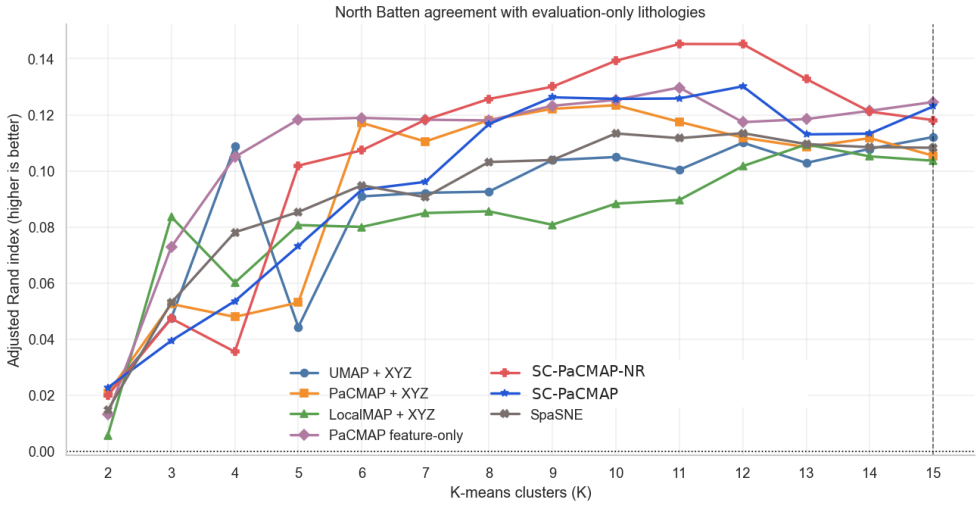
Figure 7. North Batten spatial–multivariate trade-off across $K = 2, \ldots, 15$. Lower-left is preferred; marker size increases with K, and open black rings identify $K = 15$.



(a) Spatial continuity.



(b) Multivariate delineation.



(c) Evaluation-only lithology agreement.

Figure 8. North Batten benchmark across $K = 2, \ldots, 15$. The dashed line marks the 15-class reference. Lower entropy and WCSS/TSS are better; higher ARI is better.

3.5 Cross-dataset synthesis and computational cost

The synthetic benchmark provides the strongest mechanism validation: SC-PaCMAP recovered the known repeated facies while improving both spatial continuity and feature delineation. The real datasets show the expected trade-off. Feature-only PaCMAP is the feature-compact but spatially fragmented endpoint, whereas strongly spatial solutions can be smooth but feature-poor. SC-PaCMAP consistently lies closer to the lower-left ideal, either dominating alternatives or preserving near-best feature and label agreement while delivering the strongest spatial continuity.

Observed single-fit runtimes for SC-PaCMAP in the supplied environment were 0.251, 0.736, and 1.217 seconds for the synthetic, CMiO, and North Batten datasets, respectively. These timings describe the reported workflow rather than a hardware-independent complexity claim. The deterministic benchmark establishes controlled within-workflow differences; repeated seeds and spatially blocked resampling are still required for formal uncertainty intervals.

4. Discussion

The benchmarks indicate that the principal value of SC-PaCMAP is not maximum spatial smoothness, but a better compromise between spatial continuity and multivariate delineation. These objectives are complementary in geological domaining: a partition may be geochemically compact yet spatially fragmented, or spatially smooth yet internally heterogeneous. Spatially constrained clustering makes the same trade-off explicit: greater spatial contiguity should be sought without excessively degrading homogeneity in the variables of interest (Chavent et al., 2018).

4.1 Spatial continuity as a controlled trade-off

The synthetic benchmark makes this distinction explicit. The known four facies recur in disconnected patches, so a method that treats geographic proximity as sufficient similarity can preserve the map while losing the repeated multivariate regimes. SpaSNE produced the lowest local entropy at the four-class reference but weak facies recovery (ARI = 0.1407) and poor feature delineation (WCSS/TSS = 0.9751). In contrast, SC-PaCMAP recovered the repeated facies (ARI = 0.8728) while reducing both entropy and WCSS/TSS relative to feature-only PaCMAP. This result shows that spatial regularisation is useful only when it remains conditional on attribute continuity.

The real datasets show the same trade-off under less certain geological truth. Relative to feature-only PaCMAP, SC-PaCMAP reduced entropy by 73.7% in CMiO for an 8.0% increase in WCSS/TSS, and by 59.5% in North Batten for an 8.6% increase. Feature-only PaCMAP therefore remained the most compact representation in attribute space, whereas SC-PaCMAP removed much of the spatial fragmentation at a moderate cost in within-cluster dispersion. This is the relevant result for spatial dimension reduction: the preferred solution is not the minimum of either metric separately, but a solution near the lower-left spatial–multivariate frontier. The acceptable position on that frontier depends on whether the purpose is anomaly screening, representative sampling, exploratory domaining, or subsequent spatial modelling.

4.2 Why the Phase-3 spatial attraction matters

The timing of the spatial term is central to the method. PaCMAP uses staged optimisation: mid-near pairs shape the broad arrangement early, whereas Phase 3 sets their weight to zero and focuses on local structure through neighbour attraction and further-pair repulsion (Wang et al., 2021). Introducing the spatial-continuity graph only in Phase 3 therefore allows the feature manifold and its broad relationships to form before geography modifies local structure. This is particularly important when geochemically similar material occurs in disconnected locations: applying spatial attraction too early could reorganise the embedding around physical position before the repeated feature regimes have been established.

The separate-graph formulation also preserves the complete PaCMAP feature-neighbour graph. Spatial attraction is added as a distinct objective, ramped in gradually, and funded by transferring a share of the existing nearest-neighbour attraction rather than by increasing total attraction without control. This limits abrupt deformation and reduces the risk of spatial collapse. The improvement over the earlier neighbour-replacement formulation—lower WCSS/TSS by 5.2% in CMiO and 1.4% in North Batten, together with lower entropy—supports treating feature and spatial relations as complementary graphs rather than forcing them to compete for the same fixed neighbour slots.

The global spatial strength λ provides a first level of data adaptation. It aggregates positive values of $1 - C_q$ across variance-weighted PCA components, where C_q is Geary's contiguity ratio based on squared differences between neighbouring observations (Geary, 1954). Its value was low for the synthetic data (0.125), where most dimensions were deliberately non-spatial, and higher for CMiO (0.457) and North Batten (0.511). The method consequently allocated fewer spatial edges and a smaller spatial-attraction share when the data provided weaker evidence of continuity.

However, λ should not be presented as an optimal or complete representation of spatial structure. It is a global, scale-dependent summary tied to the selected neighbourhood and sampling geometry. It is not a variogram range, anisotropy model, stationarity test, or covariance parameter. Variance weighting may also understate a low-variance geological signal and overstate broad high-variance trends. Its contribution is more modest but important: it demonstrates a practical direction in which spatial influence is estimated from the data rather than imposed through a fixed coordinate weight. Local, directional, and multiscale extensions are natural developments of this principle.

4.3 Geological interpretation and role in domaining

The CMiO result should be interpreted as strong regional organisation, not direct validation of ore types. The evaluation groups are geographically segregated, so high agreement partly reflects recovery of location. The combined evidence is more informative than ARI alone: SC-PaCMAP retained the complete feature graph, occupied a favourable trade-off region, and improved feature delineation relative to neighbour replacement. Even so, the result does not establish that each regional group is internally stationary or metallurgically homogeneous.

North Batten provides a stricter geological caution. Agreement with the 15 logged lithologies was low for all methods, while SC-PaCMAP substantially improved continuity and retained almost the same ARI as feature-only PaCMAP. This suggests that the learned structure is not a simple reconstruction of the logging taxonomy. Lithology may be incomplete as a proxy for multivariate geochemical behaviour, or the two-dimensional embedding may not preserve the distinctions represented by the logs.

For both datasets, the embedding should therefore be treated as exploratory evidence. Clusters need validation against original variables, mapping, mineralogy, texture, alteration, and, where available, metallurgical response. Recent geometallurgical work demonstrates this distinction by validating mathematically derived clusters against mineral assemblages and then propagating the interpreted classes into a three-dimensional block model (Costa et al., 2025). That level of validation is required before SC-PaCMAP clusters can be interpreted as operational geometallurgical domains.

4.4 Position relative to spatial dimension reduction

GWPCA was introduced specifically to allow covariance structure to vary geographically (Harris et al., 2011), while SpatialPCA models spatial correlation in latent factors through a location-derived kernel (Shang and Zhou, 2022). SpaSNE integrates spatial and feature information in a nonlinear visualisation objective (Zhou et al., 2025), and supervised spatial metric learning can learn response-specific spatial similarities (Zhang et al., 2025). SC-PaCMAP differs in preserving the full PaCMAP feature graph and applying a separate, unsupervised spatial-continuity graph only during local refinement.

The method also differs from coordinate concatenation. Appending standardised coordinates changes the feature metric before graph construction and makes the result sensitive to the number and scaling of all columns. In SC-PaCMAP, coordinates define only a candidate neighbourhood; the measured multivariate behaviour decides which local pairs receive spatial support. This separation is important for interpretability because geography proposes possible continuity, but does not prove it.

Spatial data are distinguished by dependence, heterogeneity, and scale, and failure to handle these properties explicitly can impair machine-learning analysis (Nikparvar and Thill, 2021). More specifically, an isotropic Euclidean metric treats equal distances alike in every direction, although spatial dependence may exhibit locally varying anisotropy (Fuglstad et al., 2015). The present method permits preferred edge directions to vary locally through candidate selection, but the geographic candidate graph remains isotropic and the PCA continuity weights remain global.

4.5 Limitations and future work

Several limitations define the next research stage. First, the reported comparisons use one deterministic embedding seed and fixed neighbourhood configurations. Exact reproduction of the PaCMAP baseline strengthens internal attribution, but it does not quantify sensitivity to initialisation, sampling geometry, or neighbourhood scale. Repeated seeds, spatially blocked resampling, and multi-dataset evaluation are needed before population-level performance claims are made (Roberts et al., 2017).

Second, global PCA and a single λ cannot represent spatial heterogeneity. GWPCA was introduced specifically to allow covariance structure to vary geographically (Harris et al., 2011), while SpatialPCA models spatial correlation in latent factors through a location-derived kernel (Shang and Zhou, 2022). A stronger extension would estimate local or domain-conditioned component weights, $\lambda(\mathbf{s})$, and use multiple graph scales to separate short-range continuity from regional trend. Such a formulation could adapt the spatial force to non-stationary areas without requiring one global compromise.

Third, the method models positive continuity but does not explicitly represent geological discontinuities. Geostatistical boundaries may be sharp (“hard”) or gradational (“soft”) (Martin and Boisvert, 2020), whereas the present graph represents both only through attraction. Faults, intrusive contacts, oxidation fronts, and hard domain boundaries may therefore require edge removal or repulsive constraints rather than weaker attraction. Future graphs should be boundary-aware and allow soft contacts, hard barriers, and repeated disconnected domains to coexist. The force allocation could also be updated during Phase 3 so that unresolved feature–spatial conflict is monitored locally rather than calibrated once at the start of the phase.

Fourth, the feature geometry inherits the assumptions of preprocessing and Euclidean distance. Geochemical and mineralogical variables may be compositional, censored, skewed, and observed over different supports. For variables that represent parts of a whole, raw Euclidean geometry can generate spurious relations; log-ratio coordinates provide a representation consistent with compositional geometry (Sadeghi et al., 2024). Future implementations should support compositional distances, detection-limit uncertainty, interval support, and variable-specific reliability within both the feature and spatial-continuity graphs.

Finally, a two-dimensional embedding can distort neighbourhoods or imply clusters absent from the high-dimensional data (Wang et al., 2021); candidate clusters are therefore not automatically stationary domains and require downstream validation (Martin and Boisvert, 2020). The current representation also does not provide domain probabilities or an inductive map for unsampled blocks. Future work should combine anisotropic and heterogeneous graph construction with ensemble uncertainty, probabilistic domain membership, out-of-sample mapping, and downstream validation against mineralogy and metallurgical response. Clustering should also be compared between SC-PaCMAP embeddings and the preprocessed original variables to test whether the embedding improves downstream partitioning by retaining meaningful structure and suppressing noise, rather than serving only as a more interpretable display (Shang and Zhou, 2022). Where reliable response labels are available, supervised spatial metric learning provides a future route to response-aware similarities and prediction (Zhang et al., 2025). In this framework, it would extend rather than replace the unsupervised spatial–multivariate representation.

Overall, the results support spatially aware dimension reduction as a useful direction for geometallurgical sampling and domaining. The contribution of SC-PaCMAP is a controlled mechanism: preserve the full nonlinear feature representation, introduce only feature-compatible spatial relations, and adapt their influence during local refinement. The benchmarks show that this strategy can reduce spatial fragmentation without reducing the embedding to a map, while the remaining limitations define a clear research path toward anisotropic, heterogeneous, multiscale, boundary-aware, and uncertainty-aware spatial embeddings.

5. Conclusion

SC-PaCMAP introduces a separate, data-adaptive spatial-continuity graph into PaCMAP while preserving the complete original feature graph. Geographic proximity supplies candidates, PCA-Geary continuity supplies evidence, and a Phase-3 budget transfer controls how strongly the spatial graph acts. The controlled synthetic benchmark showed that the method can recover repeated facies across disconnected locations rather than merely smoothing the map. CMiO and North Batten showed the expected real-data compromise: substantially lower spatial mixing with moderate feature-space cost and no requirement that external labels drive the embedding. The method should therefore be interpreted as spatial-continuity-aware exploratory dimension reduction, not automatic geological domaining or variogram modelling.

Code and data availability

The manuscript source, bibliography, figures, and citation audit accompany this draft. Before submission, the SC-PaCMAP implementation, prepared benchmark tables, executed notebooks, generated pair arrays, and environment lock should be deposited in a versioned public archive. CMiO is publicly available from Geoscience Australia (Geoscience Australia, 2021). The North Batten source releases are available through the Northern Territory Government GEMIS records (O'Rourke and Pietrass-Wong, 2015; Mortimer et al., 2018; Wallace et al., 2019). The final archive should be assigned a permanent identifier and cited in the submitted version.

Declaration of competing interest

The authors declare no competing financial or personal interests that could have influenced the work reported in this manuscript.

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A. Reproducibility checklist

The reported workflow used the following fixed settings: seed 42; two output dimensions; PaCMAP MN ratio = 0.5; FP ratio = 2.0; learning rate 1.0; 450 iterations in phases $(100, 100, 250)$; $C_{SC} = 10$; 25 Phase-3 ramp iterations; overlap factor 0.5; maximum target share 0.95; and K-means++ with 20 initializations and random state $1000 + K$. The CMiO and North Batten embeddings use $k = 10$; the synthetic benchmark uses $k = 8$. The labels and reference values of K are excluded from all embedding and allocation calculations.

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